Runhan Huang

Harvard University

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Education

Harvard University, John A. Paulson School of Engineering and Applied Sciences Aug 2026 - Now
Ph.D. in Computer Science *Cambridge, MA*

Tsinghua University, Yao Class September 2022 - June 2026
B.Eng. in Computer Science (GPA: 3.9 / 4.0) *Beijing*

Experience

MARS Lab, IIIS & Shanghai Qizhi Institute June 2024 – March 2025
Research Assistant *Beijing*

- Advised by ***Prof. Hang Zhao***, working on ***Robot Locomotion*** and ***Robot Manipulation***

Du Lab, Kempner Institute, Harvard University March 2025 – September 2025
Visiting Researcher *MA, USA*

- Advised by ***Prof. Yilun Du***, working on ***Diffusion Planning***

Publications

[P11] **Runhan Huang***, Xiangcheng Zhang*, Haoyu Chen, Yitong Li, Yen-Jen Wang, Bohan Lyu, Haotian Ye, Chuye Hong, Yixuan Wang, Haoyu Xiong, Yangzhen Wu, Enze Xie, Travis Manderson, Stefano Ermon, Pieter Abbeel, Antonio Torralba, Dawn Song, Weirui Ye, Yilun Du. *"RAX: Scaling Online Exploration for Robot Agents," arXiv*, 2026.

[P10] Haodong Yang*, **Runhan Huang***, Bowen Zhang*, Iofai Lei*, Zunzhe Zhang, Xiangcheng Zhang, Shaoting Zhu, Linzhan Mou, Yicheng Liu, Hang Zhao. *"Equilibrium Energy Control for Introspective Lifelong Robot Learning," arXiv*, 2026.

[P9] Xiangcheng Zhang*, **Runhan Huang***, Yilun Du. *"World Action Planner: Generalizable Robot Decision-Making with Action-Conditioned World Models," arXiv*, 2026.

[P8] Zhangchen Ye*, Enxuan Ruan*, Yifei Bao*, **Runhan Huang**, Jiankun Yang, Jiakang Jin, Yixiao Huo, Pengyuan Wang, Yinan Han, Huaxing Huang, Wenhao Cui, Yiming Li†, Xiaoyu Tian†. *"SkillX: Unified Multi-Skill Policy Learning for Humanoid Soccer," CoRL*, 2026.

[P7] Zunzhe Zhang*, **Runhan Huang***, Yicheng Liu, Shaoting Zhu, Linzhan Mou, Hang Zhao. *"Generative Control as Optimization: Time-Unconditional Flow Matching for Adaptive and Robust Robotic Control," NeurIPS*, 2026.

[P6] Bohan Lyu*, Yucheng Yang*, Siqiao Huang*, Jiaru Zhang*, Qixin Xu*, Xinghan Li*, Xinyang Han*, Yicheng Zhang*, Huaqing Zhang*, **Runhan Huang**, et al.. *"MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI," NeurIPS (Oral)*, 2026.

[P5] Shaoting Zhu*, Baijun Ye*, Jiaxuan Wang†, Jiakang Chen†, Ziwen Zhuang, Linzhan Mou, **Runhan Huang**, Hang Zhao. *"TTT-Parkour: Rapid Test-Time Training for Perceptive Robot Parkour," CoRL*, 2026.

[P4] **Runhan Huang**, Haldun Balim, Heng Yang, Yilun Du. *"Flexible Locomotion Learning with Diffusion Model Predictive Control," ICRA*, 2026.

[P3] **Runhan Huang***, Shaoting Zhu*, Yilun Du, Hang Zhao. *"MoELoco: Mixture of Experts for Multitask Locomotion," IROS*, 2025.

[P2] Shaoting Zhu*, Linzhan Mou*, Derun Li, Baijun Ye, **Runhan Huang**, Hang Zhao. *"VR-Robo: A Real-to-Sim-to-Real Framework for Visual Robot Navigation and Locomotion," RA-L*, 2025.

[P1] Shaoting Zhu, **Runhan Huang**, Linzhan Mou, Hang Zhao. *"Robust Robot Walker: Learning Agile Locomotion over Tiny Traps," ICRA*, 2025.

Research Experience

Generative Control as Optimization | ***ICLR Workshops 2026***

- Reinterpret flow-matching inference from fixed-schedule integration to iterative optimization by learning a time-unconditional, stationary velocity field

- Introduce an adaptive inference mechanism that dynamically allocates computation for contact-rich bottlenecks, alongside a training-free OOD anomaly detector using the vector field norm
- Scale the framework to π_0-series VLA models and demonstrate superior sample efficiency and real-world robustness on a Galaxea R1 Lite humanoid robot for high-precision tasks

TTT-Parkour: Rapid Test-Time Training for Perceptive Robot Parkour | ***CoRL 2026***

- Develop a real-to-sim-to-real framework that rapidly adapts humanoid locomotion policies to unseen and highly challenging terrain geometries through test-time training
- Build an efficient RGB-D geometry reconstruction pipeline and fine-tune a pretrained locomotion policy on high-fidelity terrain meshes in under 10 minutes
- Demonstrate robust zero-shot sim-to-real transfer across wedges, stakes, boxes, trapezoids, narrow beams, and mixed-terrain parkour courses

Flexible Locomotion Learning with Diffusion Model Predictive Control | ***ICRA 2026***

- Reinterpret diffusion models as generative priors for MPC, using reward updates, constraint projection, and candidate ranking
- Introduce a reward-weighted denoising method to finetune the diffusion planner, improving locomotion performance
- Develop real-time diffusion planning techniques such as asynchronous execution and early caching; demonstrate zero-shot transfer on a Unitree Go2 quadruped

MoELoco: Mixture of Experts for Multitask Locomotion | ***IROS 2025***

- Develop a multitask locomotion framework using a mixture-of-experts (MoE) architecture.
- Enhance sample efficiency and skill compositionality for reinforcement learning across multiple locomotion tasks.
- Project leader; contributed to model design, training infrastructure, evaluation and real-world deployment.

VR-Robo: A Real-to-Sim-to-Real Framework for Visual Robot Navigation and Locomotion | ***RA-L 2025***

- Build a digital-twin simulation pipeline using 3D Gaussian Splatting for photo-realistic rendering.
- Propose an RGB-based sim-to-real transfer policy framework for navigation and locomotion.
- Participate in low-level locomotion training and real world experiment

Robust Robot Walker: Learning Agile Locomotion over Tiny Traps | ***ICRA 2025***

- Design a proprioception-only locomotion policy that generalizes to terrains with small obstacles (”tiny traps”).
- Introduce a two-stage reinforcement learning training framework and a novel benchmark for tiny-trap traversal.
- Co-lead benchmark design, policy training, evaluation and real-robot deployment.

PROJECTS

Implicitly Exploit the Modularity in Multimodal Large Language Models | *Multimodal Machine Learning Course Project*

- Propose a new gradient update strategy for visual instruction tuning
- Train a LVLM based on phi-2 model, achieving good performance

Humanoid Perceptive Locomotion | *Intelligent System and Robotics Course Project*

- Construct a two-stage teacher–student locomotion learning paradigm with PPO and DAgger, enabling a humanoid robot to traverse challenging terrain
- Demonstrate robust stair, slope, and gap traversal in simulation using only depth image and proprioception

Multi-agent Navigation with Vision Language Model | *Practical Research Training Project*

- Build a city-scale multi-agent navigation dataset combining Google Street View, autonomous driving data, and 3D Gaussian Splatting reconstructions, enriched with landmark, orientation, and action annotations to directly supervise spatial reasoning skills.
- Design a dual benchmarking protocol spanning static VQA-style prediction and 3DGS-based simulation, providing a comprehensive assessment of perception, planning, and coordination
- Benchmark the state-of-the-art VLMs, showing that current models fall significantly short, motivating new research on embodied multi-agent intelligence

SKILLS

Programming: C, C++, Python, Pytorch, ROS, ROS2, Typescript, Scala
Robotics related software/hardware experience: isaacgym, Unitree A1, Unitree Go2, Unitree G1, Galaxea R1Lite Mobile Manipulator, PiperX Robot Arm, Franka
Key Course Taken: C/C++ Programming (A+), Calculus (A), Linear Algebra (A), General Phisics (A+), Algebra and Computation (A+), Algorithm Design, Probability and Statistic, Machine Learning, Deep Learning, Multimodal Machine Learning (A+), Computer Vision (A), Theoretical Computer Science, Reinforcement Learning, Autonomous Driving (A+), Advanced Computer Graphics (A), Practical Research Training (A)

Extracurricular experience

IIIS Soccer Team Jan 2024 – Feb 2025

Captain *Beijing*

- Led a soccer team and managed the soccer club of IIIS.